REQUIREMENTS ANALYSIS

# 🧰 Technology Stack

📅 DATE
7/9/2026

🧑‍🌾 NAME
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🌱 PROJECT
OptiCrop – Smart Agricultural Production Optimization Engine

🏆 MAX MARKS
4 Marks

## 🏛️ Technical Architecture

| COMPONENT | DESCRIPTION | TECHNOLOGY |
|---|---|---|
| User interface | Web interface for entering soil/climate data and viewing the recommendation | HTML, CSS |
| Application logic 1 | Page routing and navigation | Python (Flask) |
| Application logic 2 | Soil/climate data validation and preprocessing | Python, Pandas |
| Application logic 3 | AI-based crop recommendation | Scikit-learn, NumPy |
| User details | User enters the required soil/climate fields | Python, Flask |
| Machine learning model | Crop recommendation model | Logistic Regression, K-Means Clustering |
| Infrastructure | Application deployment on server or cloud | Local server, Render |

## 🧬 Application Characteristics

| CHARACTERISTICS | DESCRIPTION | TECHNOLOGY |
|---|---|---|
| Open source frameworks | List of the open-source frameworks used | Flask, Pandas, scikit-learn |
| Security implementations | Access controls to be used | HTTPS or local host |
| Availability | Justify the availability of the application | Local / cloud hosting |
| Performance | Design consideration for performance of the application | Pandas, NumPy, optimized ML model |